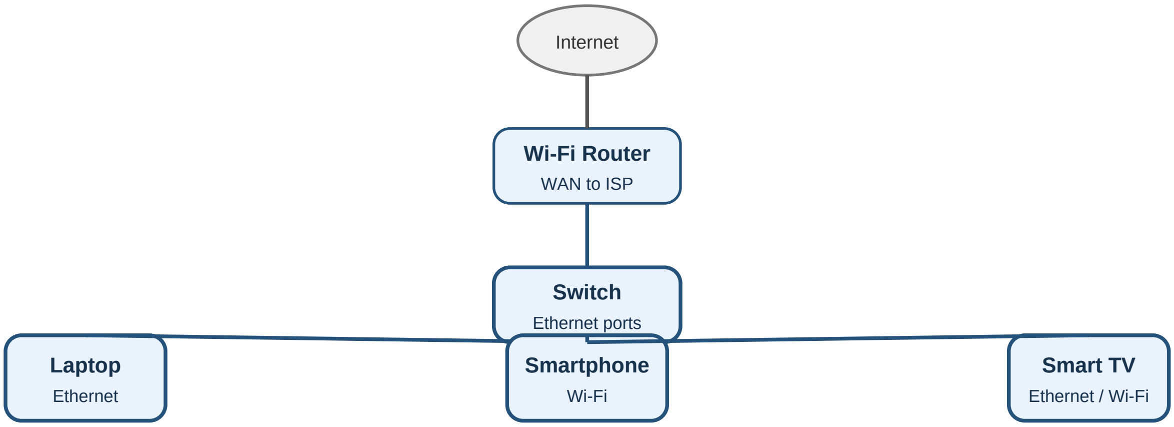
Home Network Devices Assignment

Name: ______________________________ Class: __________________

1. Home Network Diagram

The router connects the home network to the internet. A switch adds more wired Ethernet ports. Wireless devices normally connect to the router's built-in Wi-Fi access point.

Simple Home Network



Note: Smartphone usually connects to the router's Wi-Fi/access point directly; the switch serves wired devices.

2. Comparison of Network Devices

Device	What it does	Real-world use case
Router	Connects different networks and forwards traffic between the home network and the internet. It often provides Wi-Fi and DHCP.	When you watch YouTube on your phone, the router sends your internet traffic
Switch	Connects multiple wired devices in the same local network and forwards frames to the correct port.	Connect a gaming PC, smart TV, and desktop by Ethernet for stable LAN conn
Hub	Repeats incoming data to all its ports; it does not learn which device should receive it.	An older small lab might use a hub to connect test computers, but switches are
Access Point (AP)	Provides Wi-Fi so wireless devices can join a wired network.	Your phone connects to Wi-Fi to use WhatsApp, Instagram, or stream YouTube

3. How Data Travels When Ordering Food on Zomato

1. I open the Zomato app on my smartphone and select food.

2. The phone sends the request over Wi-Fi to the router's built-in access point. If a separate access point is used, it carries the wireless traffic into the wired network.

3. The router forwards the request from my home network to the internet through the internet service provider (ISP).

4. If the home network has a separate switch, the switch forwards traffic between wired devices and the router. It is not required for a phone connected directly to the router's Wi-Fi.

5. The request reaches Zomato's servers through the internet. The server sends the menu, prices, and order confirmation back along the network path to my phone.

4. Gaming PC Far from the Wi-Fi Router: Two Options

Option 1 — Powerline adapters: Plug one adapter near the router and connect it to the router with an Ethernet cable. Plug the second adapter near the gaming PC and connect it to the PC with Ethernet. The adapters carry network data through the building's electrical wiring. Performance depends on the home's electrical wiring and circuits.

Option 2 — Wi-Fi extender with an Ethernet port (or wireless bridge): Place it where it still receives a good Wi-Fi signal from the router. Connect the gaming PC's Ethernet port to the extender's Ethernet port. The extender links wirelessly to the router and gives the PC a wired Ethernet connection locally. Speed and latency depend on signal strength and extender quality.

Important: A simple Ethernet switch alone does not solve the distance problem unless you can run an Ethernet cable from the router to the switch. For gaming, a direct Ethernet cable is usually best when practical.

Conclusion: Routers connect networks, switches connect wired devices, hubs repeat traffic to all ports, and access points provide Wi-Fi. Choose a device based on distance, available cabling, speed, and reliability.